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Address

Ensenada, Baja
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Mexico

Links

Github
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Miguel Ricardo González Márquez

Data Systems Architect & Automation Lead

Summary Technical Systems Developer with a Ph.D. in Sciences, specializing in architecting robust data pipelines and automating complex workflows using Python and C++ . Proven ability to design scalable, full-stack data management dashboards powered by Node.js and Vue.js. Brings extensive experience managing large-scale datasets and optimizing system architecture. Leveraging a strong pedagogical background as a university professor, I excel at establishing rigorous technical standards, creating comprehensive documentation, and training end-users to ensure high adoption of enterprise systems.

Education

2007-2011, B.S. in Computer Systems Engineering

Instituto Tecnológico de Agua Prieta (TecNM/ITAP)

2016-2018, M.S. in Engineering

Universidad Autónoma de Baja California (UABC)

2018-2023, Ph.D. in Sciences

Universidad Autónoma de Baja California (UABC)

Professional Experience

March 2022 - Present, Lead Systems & Automation Researcher, Samsung Research Tijuana

Lead researcher architecting and optimizing large-scale data processing pipelines. Promoted to Project Lead in 2022. Responsible for structuring complex multimodal datasets, automating system workflows with Python, and ensuring strict system integrity for production-ready deployments. Serve as a core technical mentor, actively creating standard operating procedures (SOPs) and training new team members to successfully pass rigorous programming and data management skills certifications.

January 2015 - June 2016, IT Support & Network Technician, Instituto Tecnológico de Agua Prieta

Provided technical support for campus IT infrastructure; installed, maintained, and configured the campus network; developed and implemented institutional software projects.

Teaching & Mentorship Experience

August 2024 - Present, Adjunct Professor, CETYS Universidad Ensenada

Lecturer for Software Engineering and Digital Graphic Design Engineering programs. Designing and delivering project-based curricula focused on modern full-stack architectures, API integrations, and scalable front-end frameworks.

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Communication & Leadership

2025, *Panelist*, 4th Colloquium on Applications of Artificial Intelligence in Science

Roundtable panelist discussing the integration of modern data systems and AI in scientific research.

2025, *Speaker*, Samsung Week - INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "Automated Workflows and Image Generation Technologies."

2024, *Speaker*, INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "The Necessity of Lightweight Models and Optimized Pipelines."

2024, *Speaker*, DIGEEK 2024 CETYS Ensenada

Keynote presentation: "Emerging Technical and System Architectures."

2020, *Oral Presentation*, ROPEC 2020 (Ixtapa, Guerrero)

Virtual presentation at the 22nd IEEE Autumn Meeting on Power, Electronics and Computing.

2016, *Instructor*, COMEST 2016

Delivered a comprehensive Web Development course at Instituto Tecnológico de Agua Prieta for Computer Systems Engineering students.

Technical Skills

Programming Languages

- Python
- C++
- SQL
- R
- JavaScript

Web, Data & Architecture

- Node.js
- API Dev
- Vue 3 (Vite)
- MDM Principles
- Supabase (SQL)
- ETL Pipelines

Systems & Automation

- Python Scripting
- SAP ERP (Familiar)
- Workflow Auto.
- MS Excel (Adv.)
- Data Governance (Familiar)
- Documentation

Awards

2018, *Academic Merit Award*, UABC

Awarded for achieving the highest GPA in the Master's in Engineering program.

2011, *Valedictorian*, TecNM/ITAP

Awarded for achieving the highest GPA in the Computer Systems Engineering graduating class.

Interests

Professional

Passionate about bridging the gap between academia and the tech industry. Focused on data governance, system automation, full-stack workflow optimization, and mentoring the next generation of software engineers to solve real-world industrial challenges.

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Professional References

Master Elizabeth Santillán Tarazón

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Agua Prieta, Sonora.
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Master Ana Silvia Armenta Martínez

Head of Digital Graphic Design Engineering Department.
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